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Author reply

Dear Editor:

I am in complete agreement with Dr. Gandhi's comments. One of the reasons that systemic hypotension was not mentioned as a possible mechanism for ischemia to the optic nerve in association with sildenafil use in our article was the absence of microvascular risk factors in all of the five patients described. Two of the five patients had microvascular risk factors. Patient 4 had a history of nonarteritic anterior ischemic optic neuropathy (NAION) in the fellow eye. Patient 5 was a smoker and had coronary artery disease and diabetes. The other three patients did not have any identifiable microvascular risk factors. It is certainly possible that systemic or nocturnal hypotension could have been a risk factor for some or all of these patients.

Since this article was published, 10 additional cases of sildenafil-associated NAION have come to my attention. A summary of these cases is now being prepared for submission for publication. Interestingly, all of these additional 10 patients have microvascular risk factors, including hypertension, diabetes, or hyperlipidemia. Three of the patients had bilateral sequential NAION. One of them had final visual acuities of light perception (right eye) and hand motions (left eye).

I thank Dr. Gandhi for bringing the readers' attention to this important possible mechanism for sildenafil-associated NAION.

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Latanoprost versus Bimatoprost

Dear Editor:

Gandolfi and Cimino[1] report on a small series of patients who did not respond to latanoprost but did respond to bimatoprost. I congratulate them on their planned, prospective, balanced, crossover design. This design is very powerful for discerning small differences by diminishing the effects of biologic variability.

There are several items reported that need clarification. The characterization of bimatoprost as a prostamide is speculative and without literature support. The reference cited by the authors is a theoretical construct article without data.[2] Repetition of a theory does not produce a more valid statement. There is evidence that the bimatoprost free acid activates the human $F_{2\alpha}$ receptor, similar to latanoprost and travoprost.[3] There is also evidence that bimatoprost amide (Lumigan, Allergan, Inc., Irvine, CA) is hydrolyzed to the free acid by the human cornea.[4] Therefore it is more likely that bimatoprost is a prostaglandin analogue, the same as latanoprost and travoprost, but with poor corneal penetration. This would explain the need for the sixfold higher concentration of bimatoprost (0.03%) compared with latanoprost (0.005%). It is possible that some eyes are deficient in corneal esterase ability and may not be able to convert latanoprost into the free acid adequately. This is a more plausible explanation for the study findings than the prostamide theory.

The authors cite a prior work purporting to show "a greater percentage of eyes reaching target [intraocular pressures] in the low teens,"[5] yet have ignored published criticism regarding the incomplete and incorrect method of analysis to achieve this result.[6] The cited study showed no difference in intraocular pressure (IOP)–lowering ability between bimatoprost and latanoprost over the 3-month period. It is not valid to extract a subset of data for special analysis as was done for the target IOP results.

Lastly, it is difficult to generalize the findings of this study to clinical practice due to the highly selected study group. Everyone who practices medicine encounters patients who respond in unusual ways. Clinically there are patients who respond to latanoprost but do not respond to bimatoprost. What would a study of this group prove? The missing data from the present study are the denominator. How many patients were screened, in total, to find the group studied? Only by knowing the probability of finding such a patient can we make sense of the findings.

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Author reply

Dear Editor:
We read Dr. Eisenberg's letter with the utmost care. As far as the items needing clarifications, we can offer the following considerations:

1. The aim of our study was to verify if patients who did not respond to latanoprost were responders to bimatoprost. The observed results offer a clear answer to this question. Discussing the biochemistry of bimatoprost is far beyond the aims of our study. However, while discussing our results, we carefully pointed out that both (1) poor esterification of the prodrug latanoprost to the active acid form and (2) differences in the involved receptors might offer an explanation for the observed phenomenon. Moreover, the characterization of bimatoprost as a prostamide was suggested by Woodward et al in 2001.[1] Another article produced by the same authors and offering more details has recently been published.[2]
2. We were quoting the results of a multicenter, prospective, randomized clinical trial that appeared in a peer-reviewed journal.[3] This procedure is, in our opinion, appropriate and correct and follows the rules of evidence-based medicine. We are well aware of the post hoc criticisms raised by Drs. Eisenberg, Toris, and Camras in a review published, under the sponsorship of Pharmacia, in a supplement of *Survey of Ophthalmology*.[4] The issue of the face-to-face comparison between latanoprost and bimatoprost is still a matter of debate[5,6] and is not pertinent to our nonresponders study.
3. Dr. Eisenberg states that "it is difficult to generalize the findings of this study to clinical practice" and "clinically there are patients who respond to latanoprost but do not respond to bimatoprost," and asks "What would a study of this group prove?" To the best of our knowledge, our study is the first one to give proper scientific support to the general anecdotal experience quoted by Dr. Eisenberg (i.e., "everyone who practices medicine encounters patients who respond in unusual ways"). In our opinion, this is the most correct way to move from evidence-b(i)ased medicine to evidence-based medicine. Finally, the big family of the so-called hypotensive lipids (i.e., travoprost, latanoprost, bimatoprost, unoprostone) may be rather unique in the heterogeneity of the response given by each individual patient to each individual drug: does it make sense, whenever a patient gives a suboptimal response to a member of this family, to shift to another member of the same family instead of shifting to another family of drugs? Our controlled clinical trial gives some scientific support to this hypothesis.

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MRI to Differentiate Causes of Optic Nerve Disease

Dear Editor:
Rizzo et al[1] reported a retrospective evaluation of the use of magnetic resonance imaging (MRI) to differentiate optic neuritis from nonarteritic anterior ischemic optic neuropathy (NAION). They found that 5 of 32 patients with NAION had abnormally short T_1 inversion recovery signals on MRI in the clinically affected eye and that 2 patients had enhancement of the optic nerve. We have previously reported optic nerve enhancement on MRI in arteritic ischemic optic neuropathy.[2] Although it is not customary to perform an MRI for NAION, many patients arrive in the neuroophthalmology clinic with the outside MRI in hand. It has been my clinical impression that NAION does not produce enhancement of the optic nerve. When I have seen enhancement of the optic nerve in patients with a clinical picture suggestive of NAION I have pursued a more aggressive evaluation for inflammatory optic neuropathy (e.g., sarcoid) and giant cell arteritis (i.e., temporal artery biopsy).

It has been my assumption that small-vessel central nervous system ischemia (e.g., the presumed ischemic subcortical and periventricular white matter lesions) does not produce enhancement but that demyelination (e.g., acute periventricular white matter lesions in multiple sclerosis) and vasculitis (e.g., giant cell arteritis) do produce enhancement. I wonder if the authors could comment on (1) the possibility of giant cell arteritis in the two cases they reported of presumed NAION and optic nerve enhancement and (2) the mechanism for the presence (or absence) of optic nerve enhancement in NAION.

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